Band-1 Hadamard Order 4468 — Miyamoto (Paley-II seed H_1116); Gap 2052 Still OPEN

H_1116/C_1118/H_4468 PASS; 4468 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4388_Miyamoto`; `2026-09-15_Band1_Hadamard_4084_Miyamoto` (Paley-II seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4468_miyamoto/`.
Cutoff. 2026-09-15T16:30:55Z (~12:30PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1116 and 4468 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1116} ($q = 557$). Paley-I alternate for H_{1116} is **SKIP** ($q = 1115 = 5 \cdot 223$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4468 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4467 = 3 \cdot 1489$ not prime); Paley-II **FAIL** ($q = 2233 = 7 \cdot 11 \cdot 29$ not prime); Kronecker **FAIL** ($4468 = 4 \cdot 1117$, 1117 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4388 on box. This increment ships **order 4468** ($4468 = 4 \cdot 1117$, 1117 prime $\equiv 1 \pmod 4$) via Miyamoto with a direct **Paley-II seed** H_{1116} ($q = 557$). Symmetric Paley conference C_{1118} over prime 1117; $H_{4468} = \text{Miyamoto}(H_{1116}, C_{1118})$. Unit tests **41/41 PASS**. SPK next gap remains **2052** (OPEN). Order 4468 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4468; Miyamoto; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{1116} constructed and verified (Paley type II, $q = 557$). Paley-I path for H_{1116} **SKIP** ($q = 1115 = 5 \cdot 223$ not prime). 2. C_{1118} constructed and verified (Paley conference, $q = 1117$): $CC^\top = 1117I$, zero diagonal. 3. H_{4468} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $4468 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4468 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4468 = 4 \cdot 1117$
$q = 1117$	prime, $1117 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1116}
Needed conference	order $1118 = q + 1$
H_{1116} path (used)	Paley type II, $q = 557$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 1116$
H_{1116} Paley-I	**SKIP**: $q + 1 = 1116 \Rightarrow q = 1115 = 5 \cdot 223$ not prime
Conference path	Symmetric Paley over prime 1117
Paley-I for 4468	**REJECTED**: $q' = 4467 = 3 \cdot 1489$ not prime

Quantity	Value
Paley-II for 4468	**REJECTED**: $q' = 2233 = 7 \cdot 11 \cdot 29$ not prime
Kronecker for 4468	**REJECTED**: only 4×1117; $1117 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{1116}	Paley type II, $q = 557$	0	0.8006	**PASS**
H_{1116} alt	Paley type I, $q = 1115$	—	0.0	**SKIP** (not prime)
C_{1118}	Paley conference, $q = 1117$	0	1.2357	**PASS**
H_{4468}	Miyamoto(H_{1116}, C_{1118})	0	39.6384	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 42.9627 s. Compressed ± 1 matrices saved as `H1116_pm1.npz`, `C1118_pm1.npz`, `H4468_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/... + **1020** Paley-II + **1032/1092** Paley-I + **524/1048/548/1096** Double Paley-I; **exclude** 4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{1116} (Paley-II) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4468 remains a Miyamoto-only bonus.

Quantity	Value
Known count	964
4468 in SPK	**False** (Miyamoto-only)
1116 in SPK	**True** (Paley-II seed)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I/II/Kronecker/Miyamoto-Paley-conf paths for order 2052 all **FAIL** on box this run. Status **OPEN**. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN.

6. Next optional Miyamoto targets (after $q = 1117$)

q	$n = 4q$	need H	notes
1129	4516	1128	Double(PaleyI half=564)
1153	4612	1152	PaleyI(q=1151)
1229	4916	1228	PaleyII(q=613)

Also optional: method-attack SPK gap **2052**.

7. Non-claims

No Clay-closed stamp · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no finished-ToE · no invented J · no RH / P$\neq$NP / YM / Frankl / NS-solved / AC-solved claims · `clay_closed=false`. Claim level = **finite constructions ONLY**.